

17. SCAN Disk scheduling

```
# include <stdio.h>

int main () {
    int n,i,j;

    printf("Enter the number of requests: ");
    scanf("%d", &n);

    int req[n];
    printf("Enter Requests: ");
    for (i=0;i<n;i++) {
        scanf("%d", &req[i]);
    }

    int head;
    printf("Enter the the initial head posiion: ");
    scanf("%d",&head);

    int total =0;

    for (i=0;i<n;i++) {
        for (j=0;j<n-1;j++) {
            if (req[j] > req[j+1]) {
                int t=req[j];req[j]=req[j+1]; req[j+1]=t;
            }
        }
    }
    for (i=0;i<n;i++) {
        if (req[i] > head) break;
    }

    for (j=i;j<n;j++) {
        total += req[j] - head;
        head = req[j];
    }

    for (j=i-1;j>=0;j--) {
        total += head - req[j];
        head = req[j];
    }

    printf("Total seek time: %d\n",total);

    return 0;
}
```